

Scaling Purpose-Built AI Models via OCP-Aligned Distributed Inference Fabrics

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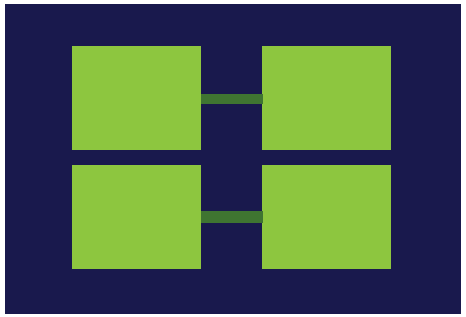


The Problem Has Shifted

Inference is no longer always a single-server problem

- Models are larger, longer-context, multimodal, and increasingly purpose-built
- Evaluate serving SLOs using tail latency—not average throughput alone
- Some deployments now need one request path to span multiple accelerator domains
- The network becomes part of the inference engine

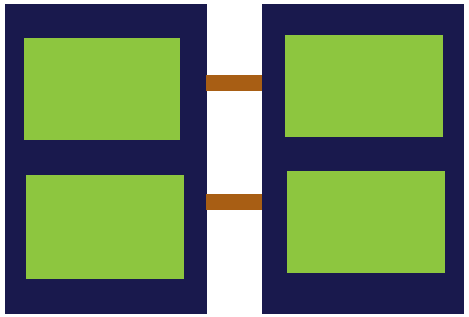
SINGLE SERVER



Local inference path

Scale-out fabric is not in the request path

MULTI-NODE SERVING



SCALE-OUT FABRIC

Fabric in request path

One request path spans multiple hosts

Bottom line: once tensors or KV state cross hosts, fabric behavior becomes service behavior.

Fast Taxonomy: Four Serving Shapes

Four deployment shapes — not maturity levels

Standalone



USE WHEN

Model and KV fit locally

FABRIC IMPLICATION

Scale-out fabric not in request path

Replicated



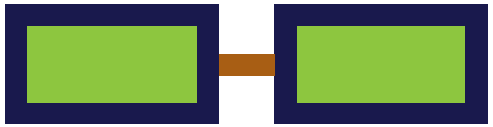
USE WHEN

Aggregate demand scales cleanly

FABRIC IMPLICATION

More north-south and storage load

Distributed



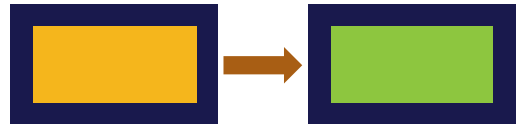
USE WHEN

One request spans systems

FABRIC IMPLICATION

Synchronous east-west path

Disaggregated



USE WHEN

Prefill/decode or roles split

FABRIC IMPLICATION

KV movement plus routing locality

Key point: these are deployment shapes, not maturity levels.

The Decision Rule

Choose the simplest serving shape that meets the SLO

SEQUENCE

1. Test the smallest viable serving unit
2. Replicate if independent requests scale cleanly
3. Distribute only when model fit or latency requires it
4. Disaggregate when phase imbalance justifies added complexity
5. Validate under burst, failure, and multi-tenant contention

END STATE

Replicated

Fabric stays outside the request path

FABRIC ENTERS THE CRITICAL PATH

Distributed

One request spans systems — synchronous east-west path

Disaggregated

Roles split — KV movement plus routing locality

Distributed inference is not the default. It is the point where the fabric enters the critical path.

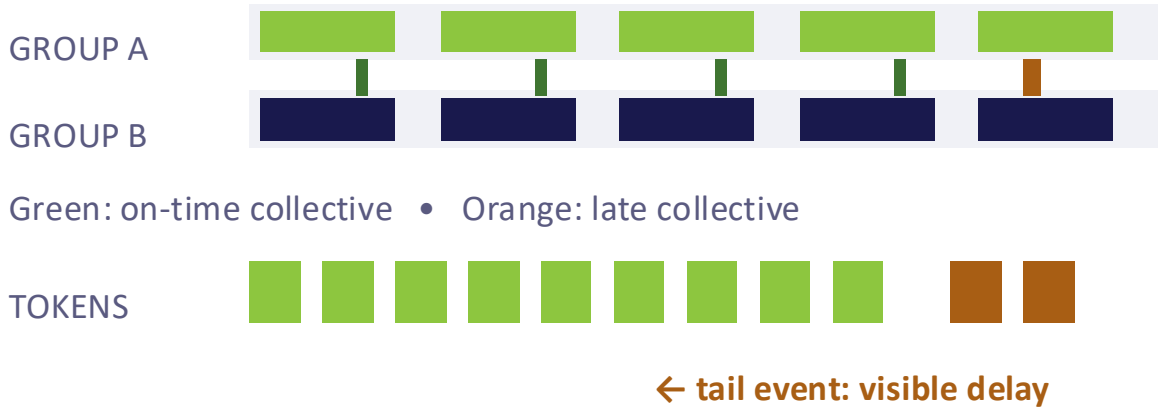
What Changes In Distributed Inference

Distributed inference turns network variance into token variance

DISTRIBUTED SERVING MAY REQUIRE

- Tensor parallel communication
- Pipeline stage handoff
- Expert routing / MoE communication
- KV-cache movement
- Cross-host synchronization
- Placement-aware scheduling

TOKEN GENERATION TIMELINE



Network variance between accelerator groups becomes variance in token completion time—including tail latency, not just the median.

The problem is not bandwidth alone. It is predictable completion time.

OCP-Aligned Fabric Building Blocks

A reusable Distributed Inference Profile

PROFILE STACK

Inference scheduler / workloads

Open Network Fabric control plane

SONiC switch operating model

Rack-scale accelerator domains · leaf-spine Ethernet

Declarative intent flows down • telemetry flows up

PHYSICAL FOUNDATION

- Rack-scale, liquid-cooled accelerator domains
- 800G / 1.6T-class scale-out interfaces
- Leaf-spine Ethernet fabric
- SONiC-based switch operating model
- Open Network Fabric control plane
- Structured cabling and plane-aware topology

OPERATIONAL FOUNDATION

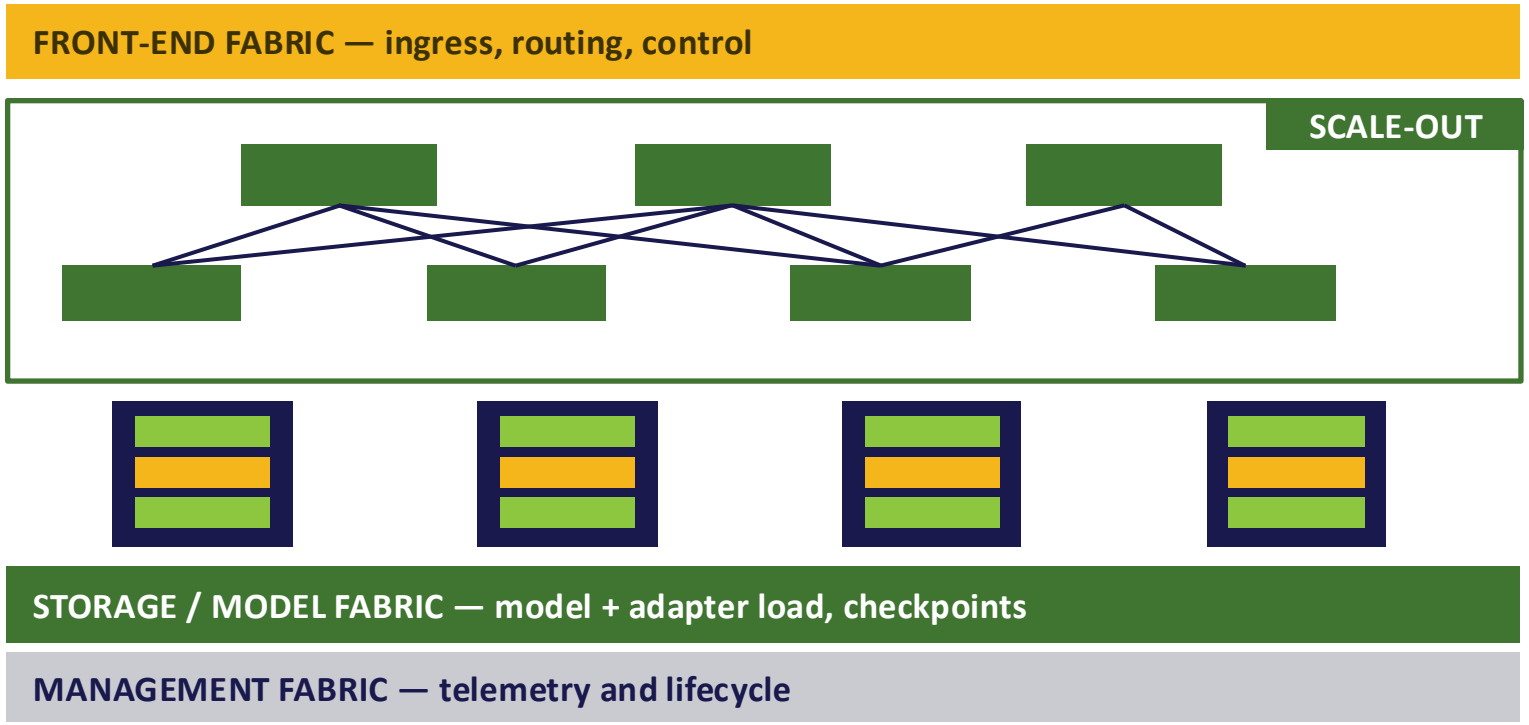
- Declarative fabric intent
- Consistent policy across switches
- Repeatable RoCE configuration
- Measurable latency and loss behavior

OCP tenets in practice: openness, scalability, and efficiency expressed as one repeatable fabric profile.

Reference Fabric View

From rack-scale domains to a distributed inference fabric

REFERENCE TOPOLOGY



TRAFFIC CLASSES

- **Scale-up domain**
  Local accelerator complex
- **Scale-out fabric**
  Cross-domain inference traffic
- **Front-end fabric**
  Ingress, routing, control, storage-facing services
- **Storage / model fabric**
  Model load, adapter load, checkpoints, datasets
- **Management fabric**
  Telemetry and lifecycle operations

Do not collapse these into one logical problem — each has a different failure and latency profile.

Fabric Profiles Without Per-Cluster Guesswork

Standardize intent and validation — not one universal queue recipe

EXAMPLE REFERENCE PROFILES — ONE PER FABRIC ROLE

Fabric role → Traffic requirements → Policy profile → Validation

SCALE-OUT / INFERENCE FABRIC

PURPOSE

Synchronous accelerator communication

Tensor / expert parallel · KV transfer

EXAMPLE PROFILE

- RDMA-aware transport
- PFC + ECN where required
- Low-loss / bounded-queue design
- Consistent ECMP behavior
- Tail-latency validation

LATENCY + JITTER CRITICAL

STORAGE / MODEL FABRIC

PURPOSE

Model + adapter load

Checkpoints · datasets · GPU storage I/O

EXAMPLE PROFILE

- TCP or RoCE/RDMA — workload dependent
- Dedicated RoCE is a first-class design
- Congestion policy matched to storage transport
- Burst + throughput validation

TRANSPORT DEPENDENT

FRONT-END / AI SERVICES FABRIC

PURPOSE

Inference APIs · gateways

Agent runtimes · RAG services

EXAMPLE PROFILE

- TCP / HTTP / gRPC
- SLO-driven QoS where needed
- Protect critical service paths
- Validate burst + tail latency

SERVICE-SLO DRIVEN

MANAGEMENT / CONTROL FABRIC

PURPOSE

Telemetry · lifecycle · administration

Infrastructure control traffic

EXAMPLE PROFILE

- Best effort for routine management
- Protect critical control-plane traffic
- Control-plane policing / queue protection as appropriate
- Validate reachability under failure

AVAILABILITY CRITICAL

DEDICATED OR CONVERGED?

Dedicated fabrics

Each role receives the policy appropriate to its endpoints, transport, and SLO.

Converged fabrics

The profile additionally defines traffic-class isolation, QoS, congestion interactions, and validation for workloads sharing links.

REFERENCE PROFILE ≠ UNIVERSAL CONFIGURATION

Exact queueing, PFC, ECN, buffer, MTU, and congestion-control settings depend on NICs, switches, transports, topology, endpoints, and workload SLOs. Profiles should specify supported combinations and validation tests.

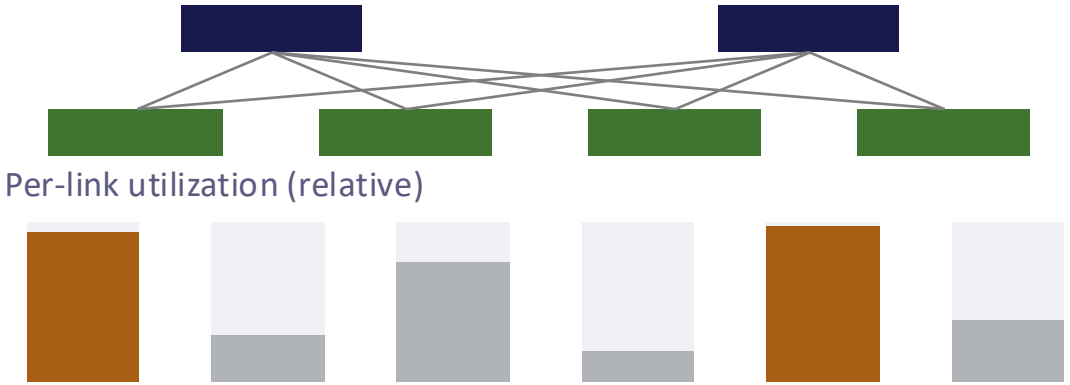
Standardize the policy model and validation method; tune each fabric role to the traffic it actually carries.

ECMP Path Selection And Tail Latency

Available paths only help when the workload can use them

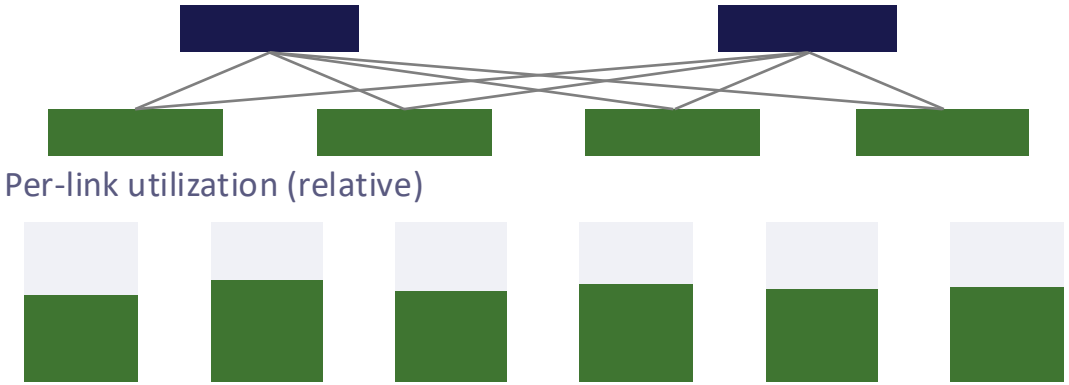
Traffic concentrates on a subset of paths → **Hotspots and queueing develop** → **Communication stragglers delay synchronized work** → **Inference SLOs are at risk**

PATH CONCENTRATION



Some links congest while other capacity remains available

VALIDATED PATH DISTRIBUTION



Workload-aware validation of available path utilization

Illustrative traffic distribution — not benchmark results.

Validate flow entropy and path distribution; avoid cross-stage hash polarization.

Measure completion-time tails and inference SLOs under contention and failure.

Disaggregated Inference: Useful, But Not A Standard Topology

Disaggregation separates roles; the fabric must preserve locality

COMMON EMERGING PATTERN



Assigns work by load, cache, and locality

Process prompt context

Generate output tokens

KV state may move across workers, nodes, or racks.

DESIGN CAUTION

- Disaggregation is not automatically higher throughput
- Implementations differ across serving stacks
- Treat it as a placement and traffic pattern, not a universal architecture

EXAMPLE LOCALITY SCOPES

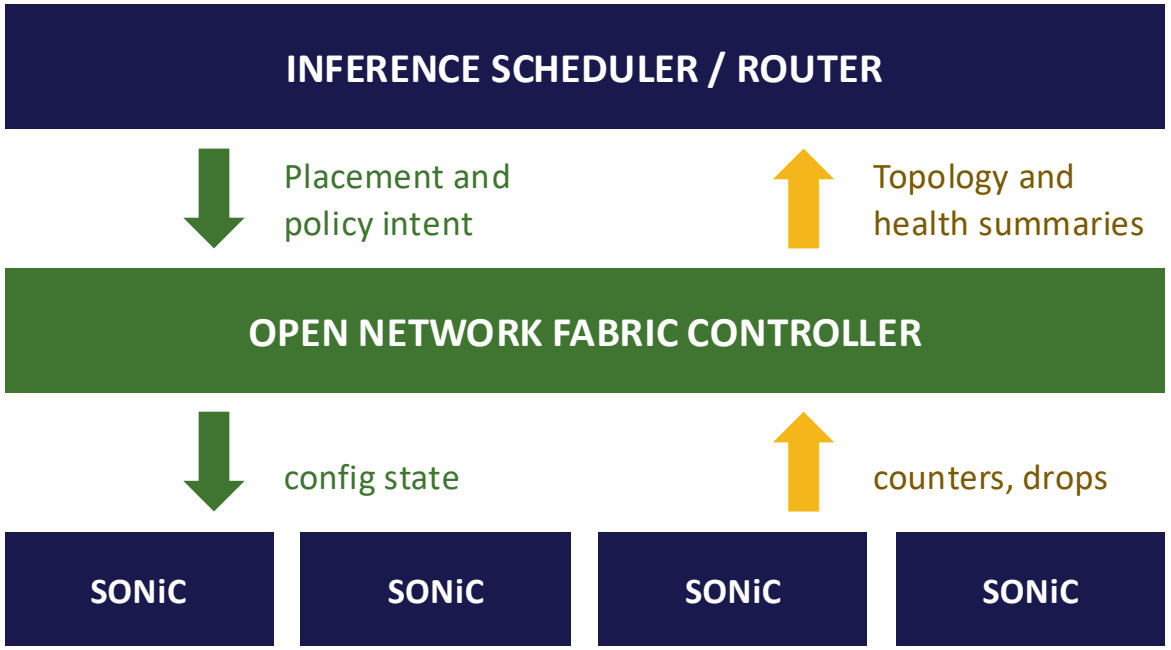
- Same host
- Same leaf domain
- Same fabric pod
- Cross-pod

Locality informs transfer cost—it does not determine it. Validate data volume, usable bandwidth, endpoint locality, and contention.

Why Open Fabric Control Matters

The serving stack and the network need a shared operating model

PROPOSED SERVING–FABRIC INTEGRATION



One auditable control plane across every switch.

A PRODUCTION FABRIC SHOULD EXPOSE

- Topology and locality information
- Traffic-class policy
- Congestion and drop telemetry
- Link and path health
- Repeatable configuration state
- Workload-aware validation hooks

Without these, fabric health is an assertion. With them, it is a measurement the serving stack can act on.

Inform placement and admission; keep fast congestion control in NICs, switches, and transport.

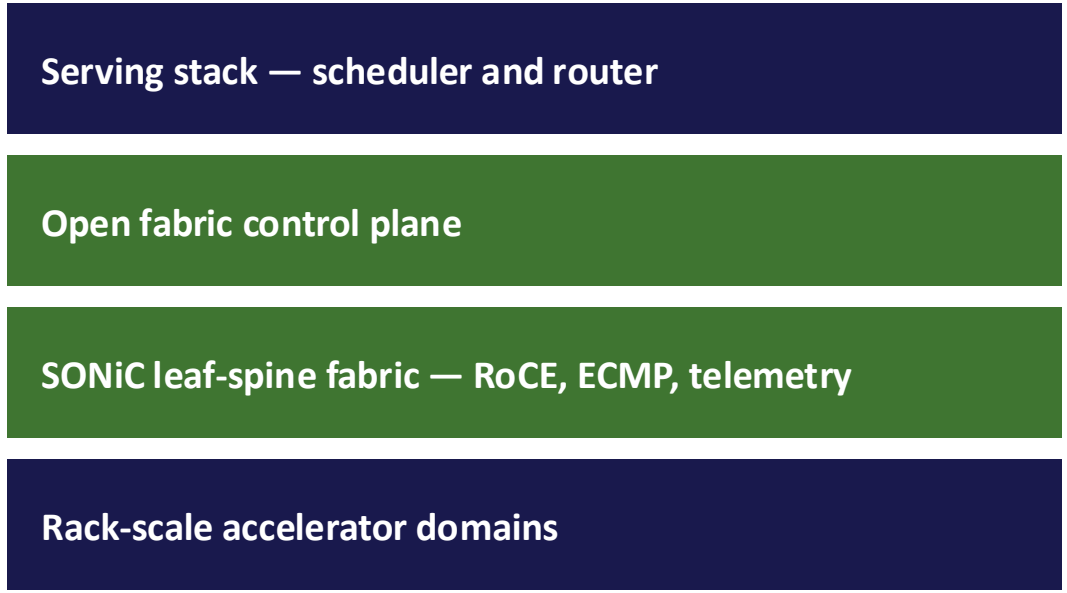
Closing Architecture Principle

Distributed inference succeeds when the fabric is designed as part of the serving system

TAKEAWAYS

- Start with the smallest measured serving shape
- Use distributed inference only when the model or SLO requires it
- Treat RoCE, ECMP, and topology locality as service-level controls
- Standardize the fabric profile so deployments are repeatable
- Keep the architecture open: OCP-aligned hardware, SONiC, and open control

REFERENCE PROFILE



Predictable inference needs predictable fabric behavior.

Call to Action

- Community objective
  - Make fabric behavior repeatable, measurable, and comparable across inference deployments.
- Proposed contribution: an OCP Distributed Inference Fabric Profile
  - Fabric-role policies, multipath validation, and workload-based SLO tests.
- How to get involved
  - Open Network Fabric and SONiC project communities
- What we are asking the community for
  - Reference topologies and measured p99 behaviour — not bandwidth numbers alone
- Additional information
  - https://www.opencompute.org/projects

Thank You!

